1. Functions

- A function assigns a collection of x values to **only one** y value.
  - A function can be denoted by $f(x)$, $g(x)$, $h(x)$ etc.
- Inputting a x value into the function f outputs a y value [also written as $f(x)$]

For example

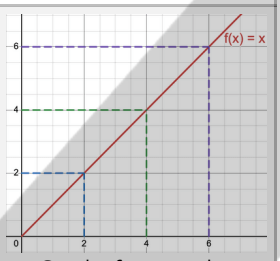
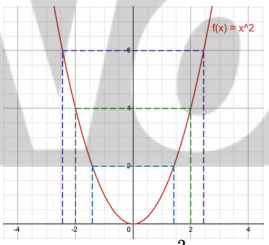
$$f(x) = x^2 + 4x + 4$$

When $x = 2$

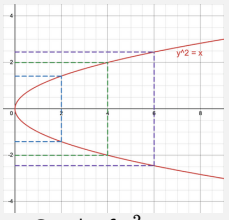
$$f(2) = (2)^2 + 4(2) + 4 = 16$$

1.1. Types of functions and relations

There are many types of relations a function can have:

Function	Definition	Graphical representation
One to One	**One** x-value **input** gives one y-value **output**. As an example, all linear functions of the type $f(x) = mx + c$ are One to One.	 Graph of $y = x$, also written as $f(x) = x$.
Many to One	**Two** x-value **inputs** give the same y-value **output**.	 Graph of $y = x^2$, also written as $f(x) = x^2$.

Another type of relation, that is **not a function** is:

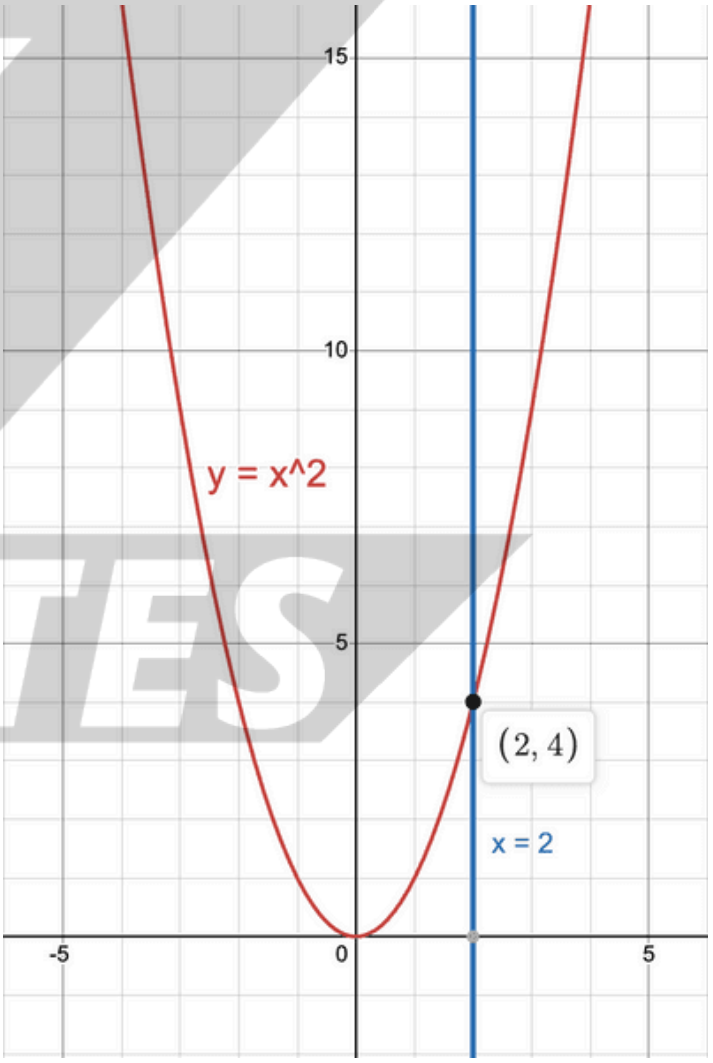
Relation	Definition	Graphical representation
One to many [Not a function]	**One** x-value **input** gives **two** y-value **outputs**. This is **not a function** as it gives us more than one y-value output with one x-value input, contradicting the definition of a function.	 Graph of $y^2 = x$.

Vertical Line Test to check for Existence of a Function

To check for the existence of a function, we make use of the vertical line test.

- If a vertical line, $x = a$, cuts the the function at **one point only** then the function exists
  - Note that a is a constant which is included within the interval of x-values the function is defined for.

Curve passes the vertical line test



- As the curve $y = x^2$ passes the vertical line test, it is a function.

Curve fails the vertical line test

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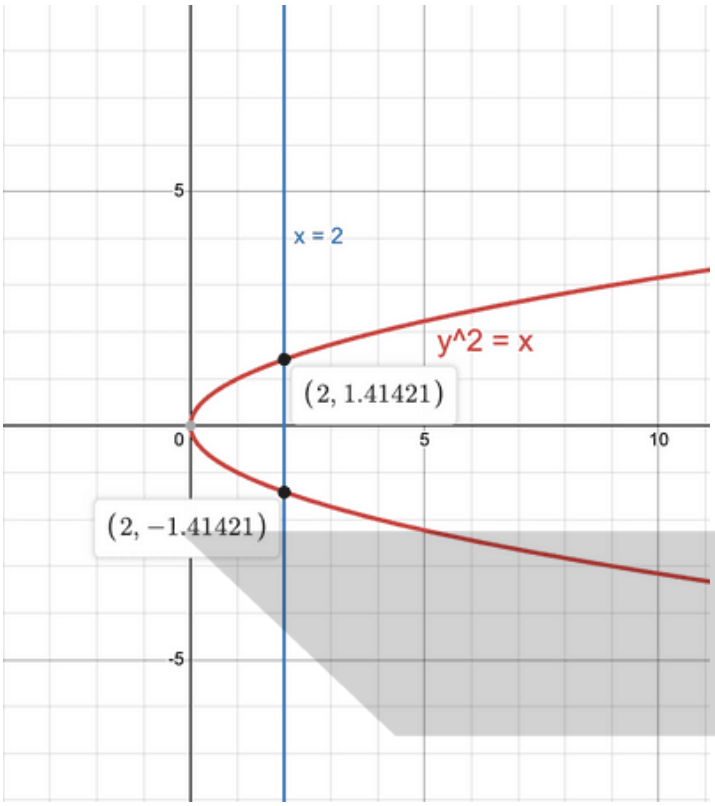


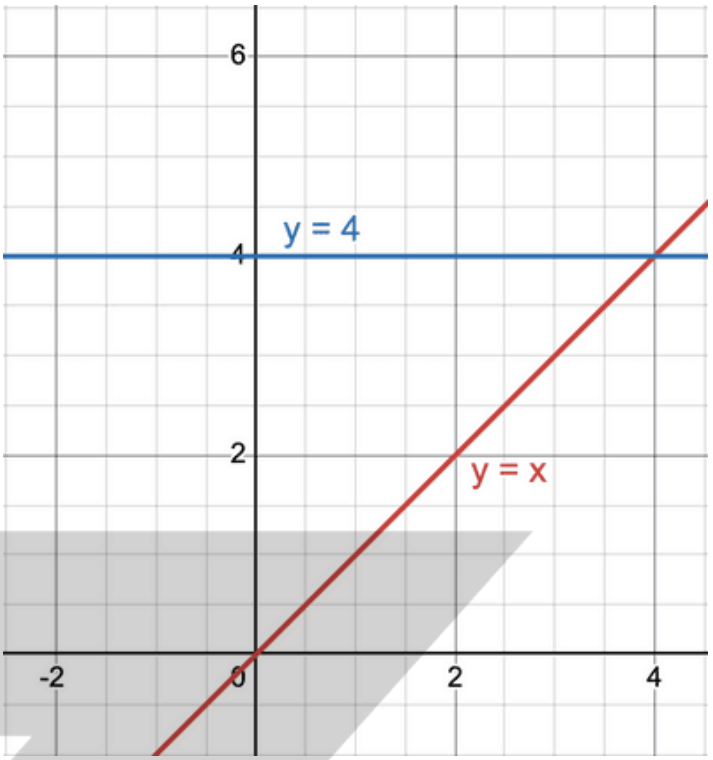
- As the curve $y^2 = x$ fails the vertical line test, it is not a function.

Horizontal Line Test to check for One to One Relation

To check if a function has a One to One relation, we make use of the horizontal line test.

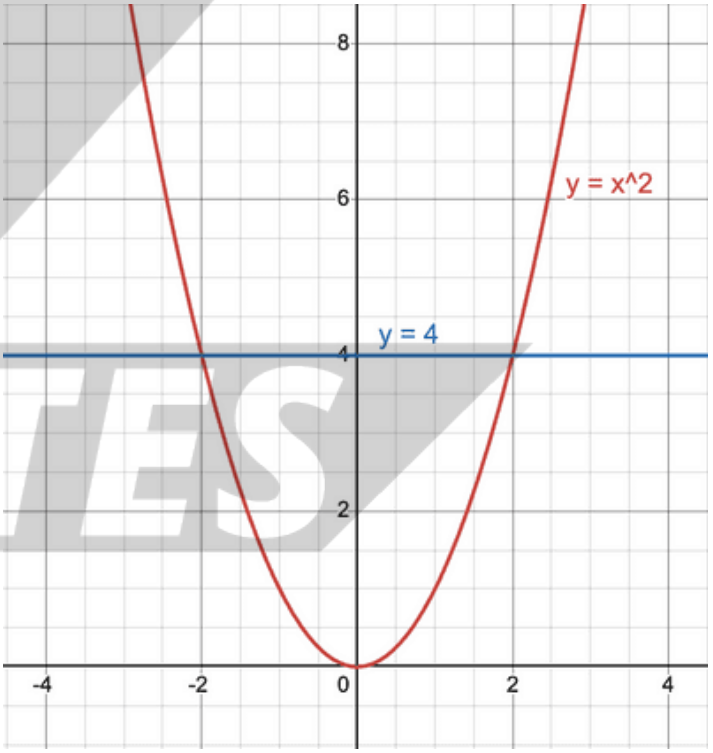
- If a horizontal line, $y = a$, cuts the function at **one point only** then the function has a One to One relation. Else it will have a Many to One relation.
  - Note that a is a constant which is included within the interval of y-values the function is defined for.

Curve passes the horizontal line test



- As the curve $y = x$ passes the horizontal line test, the function has a One to One relation.

Curve fails the horizontal line test



- As the curve $y = x^2$ fails the horizontal line test, the function does not have a One to One relation. It instead has a Many to One relation in this case.

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1.2. Domain and Range

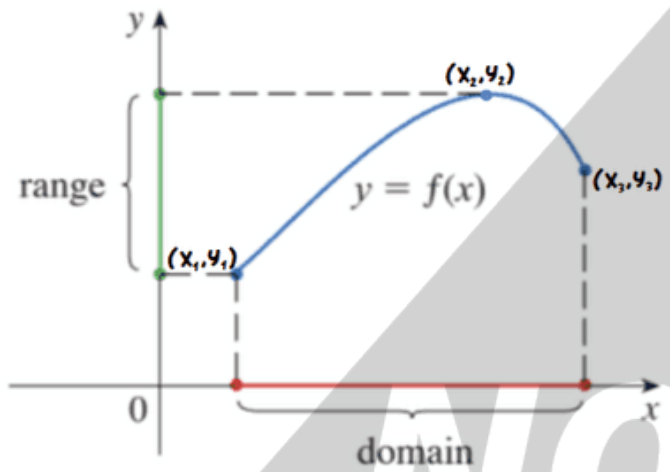
Domain

- The **domain** of a function is the set of x-values, or **inputs**, that the function f is defined for.
  - It consists of the **smallest** and **biggest** x-value the function accepts as an input, or is defined for.
  - Example: $x \in \mathbb{R}$ implies the function the function is defined for all real values of x.
  - $4 < x < 10$ implies the function is defined between x = 4 and $x = 10$.

Range

- The **range** of a function is the set of corresponding y-values, or **outputs**, that the function f is defined for.
  - It consists of the **lowest** and the **highest** y-value the function outputs for a given domain.
  - Example: $y \in \mathbb{R}$ implies the function is defined for all real values of y.
  - $-2 \leq y \leq 5$ implies the function is defined between $y = -2$ and $y = 5$, where both values are inclusive.

Example



For this curve, $y = f(x)$:

- The range is $y_1 \leq f(x) \leq y_2$
  - Also written as $y_1 \leq y \leq y_2$
- The domain is $x_1 \leq x \leq x_3$

1.3. Domain and Range of a Quadratic

Domain of a Quadratic

- The domain of a quadratic is $x \in \mathbb{R}$, unless it is restricted.
- The line $x = -\frac{b}{2a}$, or $x = q$ (where q is the x coordinate of the vertex), is the line of symmetry
  - We can find a restricted domain such that the function has a One to One relation.

Range of a Quadratic

- The range of a quadratic depends on its minimum or maximum y value.

Consider a quadratic equation written in the form

$$ax^2 + bx + c = 0$$

We can complete the square for this equation to get the y-coordinate of the vertex, or the minimum or maximum y value.

Equation	Coordinates of vertex	Nature of vertex	Range
$y = p(x - q)^2 + r$	(q, r)	Maximum point ($a < 0$ or $p < 0$)	$r \geq y$ or $r > y$
$y = p(x - q)^2 + r$	(q, r)	Minimum point ($a > 0$ or $p > 0$)	$r \leq y$ or $r < y$

- Note that the type of inequality sign you use ($>$ or $\leq$) depends on if the domain includes the x coordinate of the vertex, q, using $\leq$ and $\geq$.

Example 1

Given a quadratic, $f(x) = 2x^2 - 2x + 3$ with a domain of $a \leq x$, find the smallest value of a for which the function has a One to One relation.

Answer

First we complete the square for $f(x)$

$$2\left(x - \frac{1}{2}\right)^2 + \frac{5}{2}$$

As $a > 0$ and $p > 0$, the vertex is a minimum point with coordinates $\left(\frac{1}{2}, \frac{5}{2}\right)$.

So the smallest value of a, for which the function has a One to One relation, is $\frac{1}{2}$ and the domain is

$$\frac{1}{2} \leq x$$

The graph is shown below

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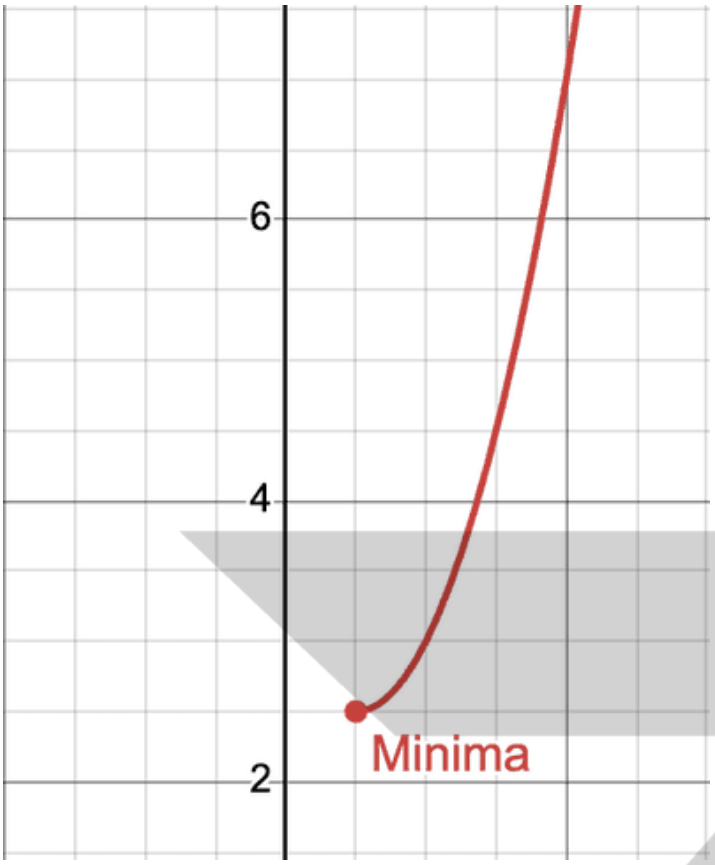


Example 2

Find the range of $f(x) = -2x^2 + x + 4$, where $x \in \mathbb{R}$.

Answer

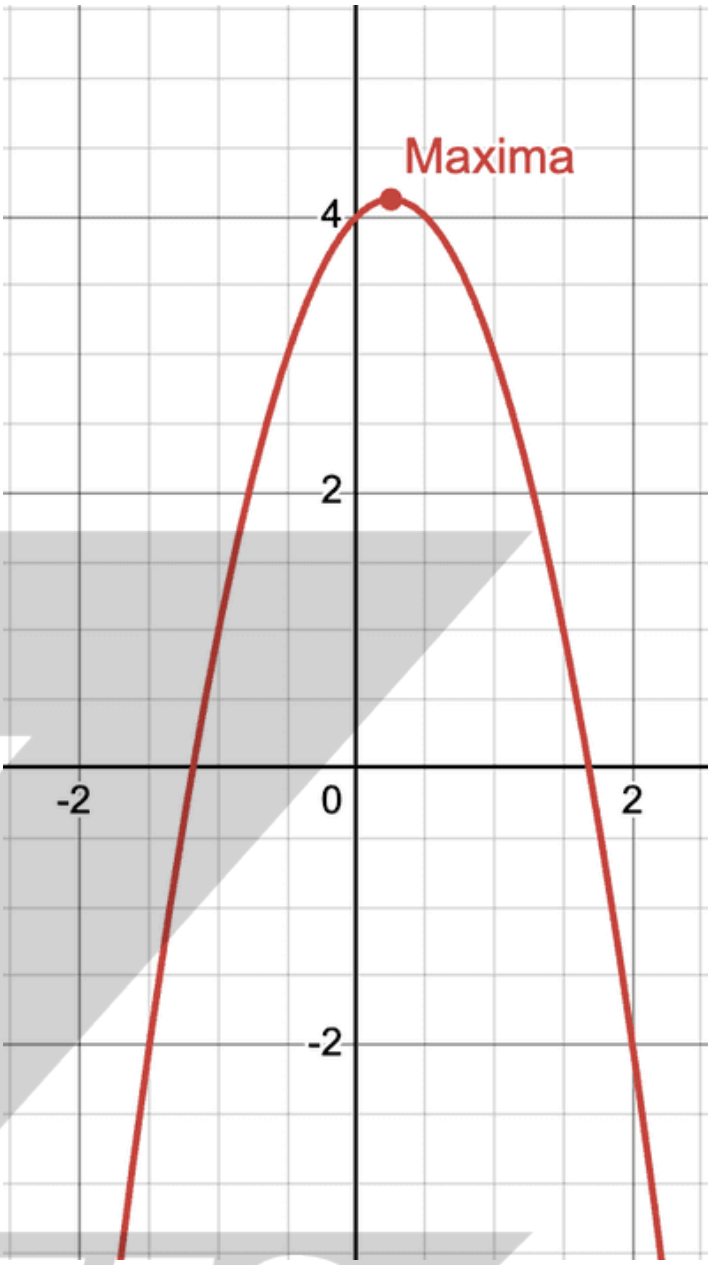
First we complete the square for $f(x)$

$$f(x) = -2\left(x - \frac{1}{4}\right)^2 + \frac{33}{8}$$

As $a < 0$ and $p < 0$, the vertex is a maximum point. We can now find the range as

$$\frac{33}{8} \geq y$$

The graph is shown below



1.4. Different types of Functions

Square root function

- Consider a function $f(x) = \sqrt{ax + b}$
  - The **domain** must be chosen such that $ax + b > 0$ always. In this case the domain is $x > \frac{-b}{a}$.
  - The **range** of such a function is always $y > 0$
- For a more general way of solving, consider a function $f(x) = \sqrt{g(x)}$ [where $g(x)$ is any function such as a linear function, quadratic, or even a higher degree polynomial]
  - The **domain** must be chosen such that $g(x) \geq 0$ always
  - The **range** of function is always a positive, non-zero, interval.

Rational function

- A rational function is a function written in the form $\frac{f(x)}{g(x)}$, where $g(x) \neq 0$.

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- To find the domain
  - Find the values of x for which $g(x) = 0$. These values of x cannot be inputted into the function and must be excluded in the domain.
- To find the range
  - Input a large x value, such as 999 and -999 . This will help you find the y value which the rational function is approaching, but never reaches. This limits the range of the rational function.

If the function is written in the form

$$f(x) = a + \frac{g(x)}{h(x)} \text{ for } x \in \mathbb{R}$$

Where a is a non-zero constant, and $g(x)$ and $h(x)$ are functions.

- Then the **range** is $f(x) < a$ or $f(x) > a$ depending on if the function is approaching the value a from above or below the line $y = a$. This can be checked by inputting a large x value and seeing if the output is greater than or less than a.
  - Note that the inequality can change if the domain changes.

Example

Find the range of

$$f(x) = \frac{2}{3} + \frac{2}{3(3x-1)} \text{ for } x > \frac{1}{3}$$

Answer

Inputting in a large x value, such as 999, gives us

$$f(999) = \frac{2}{3} + \frac{2}{3(3(999)-1)}$$

Which is greater than $\frac{2}{3}$, so the function is approaching this value from above. This gives us the range of the function as

$$y > \frac{2}{3}$$

1.5. Inverse Function

- An inverse function reverses another function.
  - Commonly denoted by $f^{-1}(x)$.
  - It is also important to note that $f^{-1}(x)$ does **not** imply $\frac{1}{f(x)}$.
- Inputting x into the function f outputs $f(x)$ while inputting $f(x)$ into f^{-1} outputs x [also written as $f^{-1}(x)$]

Consider a function f and its inverse function f^{-1} . These two functions can be represented as

And

This gives us an important result:

$$f^{-1}(\, f(x)\,) = f(\, f^{-1}(x)\,) = x$$

- An inverse function, f^{-1} , can **only** exist if the function f has a **One to One** relation.
  - To check if an inverse function exists, we can use the **horizontal line test**.

Domain and Range of an Inverse Function

Using the previous result, $f^{-1}(\, f(x)\,) = f(\, f^{-1}(x)\,) = x$, we can find the domain and range of an inverse function:

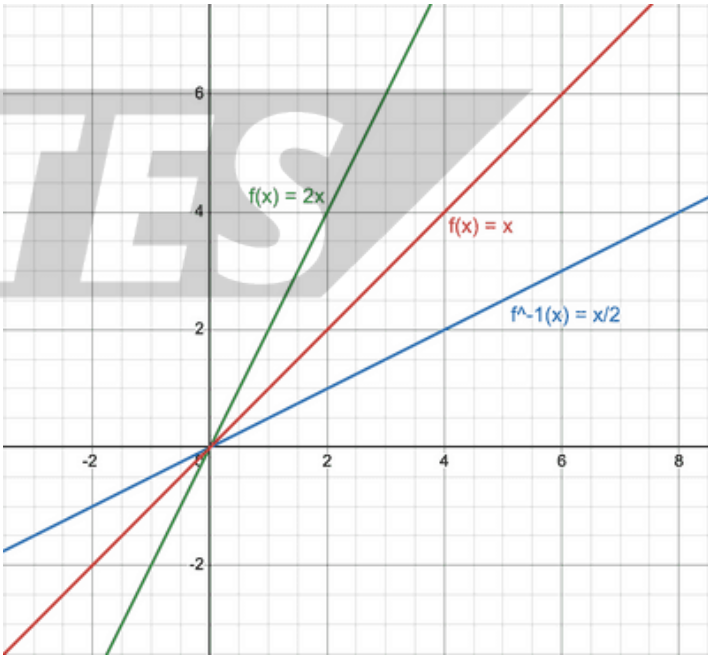
Function	Domain	Range
$f(x)$	$a < x < b$	$c < y < d$
$f^{-1}(x)$	$c < x < d$	$a < y < b$

We observe:

- The domain of $f(x)$ is the range of $f^{-1}(x)$, and vice versa
- The range of $f(x)$ is the domain of $f^{-1}(x)$, and vice versa.

Relationship between a Function and its Inverse

Using the domain and range of a function and its inverse, we can graph both functions



Observations:

- Both functions are **symmetric** to the line $f(x) = x$
- Both functions **only** intersect at $f(x) = x$

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Finding an Inverse Function

- We can find the equation for an inverse function by "swapping" all the x variables to y, and y variables to x, and then making y the subject of the formula again.

Example

Given $f(x) = 2x + 3$, find the inverse of this function.

Answer

We start by swapping the x and y values

$$y = 2x + 3 \implies x = 2y + 3$$

Now we can make y the subject of the formula

$$y = \frac{x-3}{2}$$

So we get the inverse function as

$$f^{-1}(x) = \frac{x-3}{2}$$

Example 1

Given

$$f(x) = (x-3)^2 - 2 \text{ for } x > 4$$

Find an expression for $f^{-1}(x)$ and find the domain of f^{-1} .

Answer

We start by swapping the x and y values.

$$y = (x-3)^2 - 2 \implies x = (y-3)^2 - 2$$

Now we can make y the subject of the formula

$$y = \pm\sqrt{x+2} + 3$$

- To correctly pick between + and - for the square root, we look at the domain of f. As the domain of f is $x > 4$, the range of f^{-1} must be $y > 4$. Due to this, we need to pick the positive value of the square root, as it is the only way we will get a y value output greater than 4.

So we get the inverse function as

$$f^{-1}(x) = +\sqrt{x+2} + 3$$

We can find the domain using the fact that the range of f is the domain of f^{-1} .

At $x = 4$

$$f(4) = (4-3)^2 - 2 = -1$$

At $x = 5$

$$f(5) = (5-3)^2 - 2 = 2$$

The y value ouput of the quadratic increases as the x value input increases as well. So the range of f is

$$f(x) > -1$$

This gives us the domain of f^{-1} as

$$x > -1$$

Example 2

Given

$$f(x) = -(x-3)^2 + 4 \text{ for } x \geq m$$

Find the smallest value of m such that the inverse function f^{-1} exists.

Answer

- An inverse function, f^{-1} , can **only** exist if the function f has a **One to One** relation.

So we need to find the value of m such that $f(x)$ is an One to One function.

We can deduce that the coordinates of the vertex are $(3, 4)$ from the given equation. Using this, we get the smallest value of m as 3.

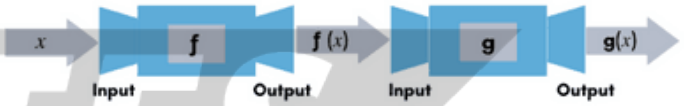
So the domain for which the inverse function exists is

$$x \geq 3$$

1.6. Composite Function

- A composite function is a combination of two functions
  - Commonly denoted by $g(\,f(x)\,)$ or $gf(x)$.
- Inputting a x value into the function f outputs $f(x)$. The same value, $f(x)$, is then inputted into another function g which outputs $g(x)$.

A summary of the process has been described in the image



- It is also important to note that $g(\,f(x)\,)$ does not always equal $f(\,g(x)\,)$
- $f^2(x)$ implies $f(\,f(x)\,)$, unless stated otherwise.

Domain and Range of a Composite Function

- As $f(x)$, or the y-value output of the function f, is inputted into the function g as a x-value input
  - The domain and range of g depends on the range of f.
- If the interval of **range values** for $f(x)$ is not included in the interval of **domain values** for $g(x)$, the composite function **cannot exist.**
  - If the range of $f(x)$ is included in the domain of $g(x)$, but is a smaller interval, then the domain of the composite function becomes a smaller interval too.

Example 1

Given

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$$f(x) = 2(x+3)^2 - 7 \text{ for } x \leq -4$$

$$g(x) = 2x - 3 \text{ for } x \leq k$$

Find the largest value of k for which the composite function $f(\,g(x)\,)$ is defined.

Answer

- For a composite function $f(\,g(x)\,)$ to exist, the range of $g(x)$ must be included in the domain of $f(x)$.

The domain of $f(x)$ is $x \leq -4$, so we must find a value of k for which the range of $g(x)$ is $y \leq -4$

Letting $g(x) = -4$, we get

$$2x - 3 = -4 \implies x = -\frac{1}{2}$$

So the largest value of k is

$$k = -\frac{1}{2}$$

Example 2

Given

$$f(x) = -2x^2 + 12x - 3 \text{ for } x \in \mathbb{R}$$

$$g(x) = 2x + 5 \text{ for } x \in \mathbb{R}$$

Find the values of x for which $g(\,f(x)\,) + 1 = 0$.

Answer

We can find the function $g(\,f(x)\,)$ by inputting $f(x)$ into $g(x)$

$$g(\,f(x)\,) = 2(-2x^2 + 12x - 3) + 5 \implies -4x^2 + 24x - 1$$

Now we can input this value into the equation we need to solve, $f(\,g(x)\,) + 1 = 0$

$$-4x^2 + 24x - 1 + 1 = 0 \implies 4x(-x + 6) = 0$$

So the values of x which satisfy the equation are

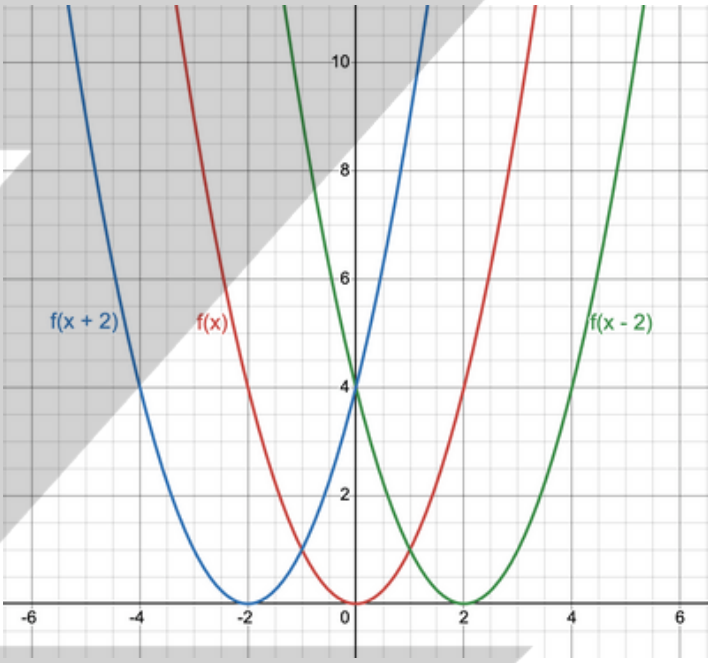
$$x = 0 \text{ or } x = 6$$

1.7. Translations

- A translation moves all the points on a curve in the x or y direction, depending on the translation.

Function	Transformation	Result	Geometric interpretation	New points
$f(x)$	$\binom{a}{0}$	$f(x-a)$	Translation of a units in the **rightward**, or **positive**, x direction.	$(x+a, y)$
$f(x)$	$\binom{-a}{0}$	$f(x+a)$	Translation of a units in the **leftward**, or **negative**, x direction.	$(x-a, y)$
$f(x)$	$\binom{0}{b}$	$f(x)+b$	Translation of b units in the **upwards**, or **positive**, y direction.	$(x, y+b)$
$f(x)$	$\binom{0}{-b}$	$f(x)-b$	Translation of b units in the **downwards**, or **negative**, y direction.	$(x, y-b)$

x direction translation



y direction translation

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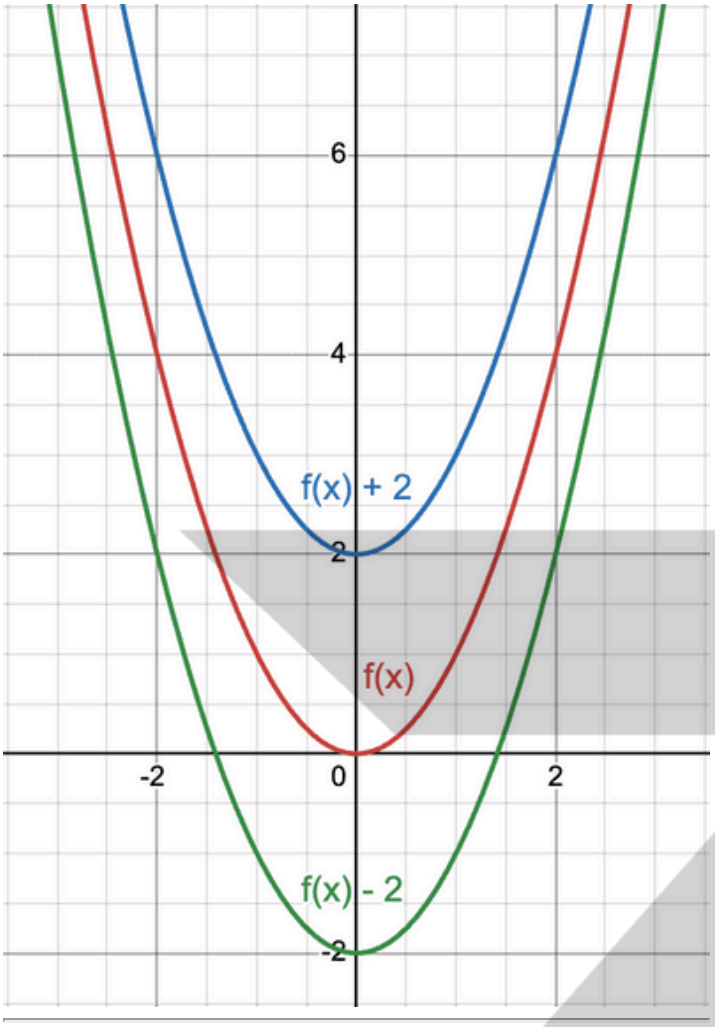


Example

The graph of $f(x) = x^2 + 3x + 2$ has been translated by the vector $\binom{-1}{3}$. Find the equation of the resulting graph.

Answer

Applying the translation to the function $f(x)$, we get:

$$y = f(x+1) + 3$$

$$y = (x+1)^2 + 3(x+1) + 2 + 3$$

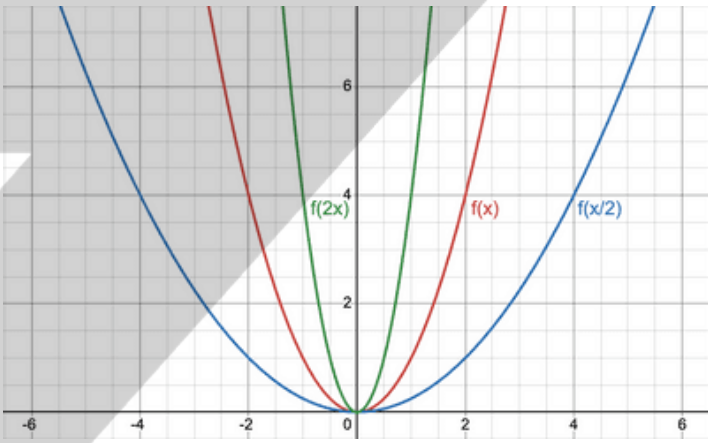
$$y = x^2 + 5x + 9$$

1.8. Stretches

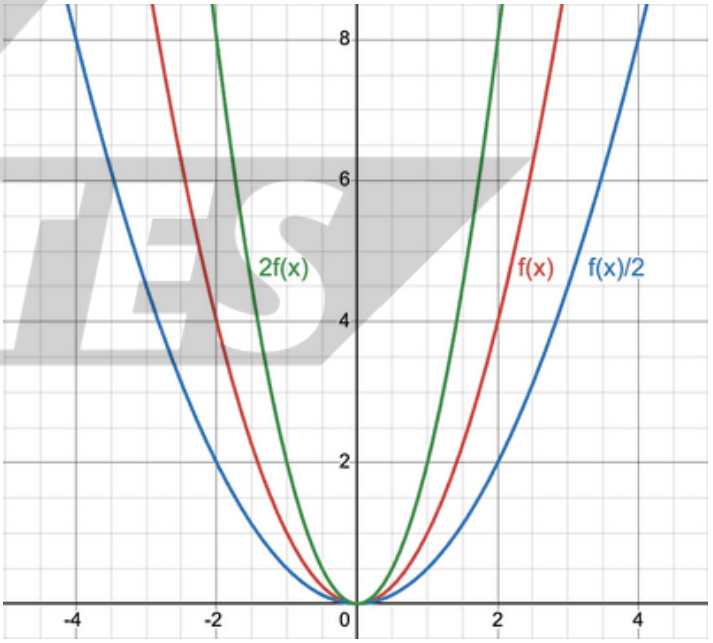
- A stretch multiples all the x or y points on a curve by the stretch factor a.

Function	Stretch factor and direction	Result	Geometric interpretation	New points
$f(x)$	Stretch in the x-axis by factor of $\frac{1}{a}$	$f(ax)$	Stretch parallel to the x-axis by a scale factor of $\frac{1}{a}$	$\left(\frac{x}{a}, y\right)$
$f(x)$	Stretch in the x-axis by factor of a	$f\left(\frac{x}{a}\right)$	Stretch parallel to the x-axis by a scale factor of a	(ax, y)
$f(x)$	Stretch in the y-axis by a factor of $\frac{1}{a}$	$\frac{f(x)}{a}$	Stretch parallel to the y-axis by a scale factor of $\frac{1}{a}$	$\left(x, \frac{y}{a}\right)$
$f(x)$	Stretch in the y-axis by a factor of a	$af(x)$	Stretch parallel to the y-axis by a scale factor of a	(x, ay)

Stretch parallel to x-axis



Stretch parallel to y-axis



Example 1

The graph of $f(x) = 5x^2 + 2$ is stretched by a factor of 2 along the y-axis, find the resulting equation of the graph.

Answer

for this stretch factor

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$$y = 2f(x)$$

$$y = 2(5x^2 + 2)$$

$$y = 10x^2 + 4$$

Example 2

The graph of $y = x^2 + 3x + 2$ is stretched by a factor of $\frac{1}{3}$ along the x-axis, find the resulting equation of the graph.

Answer

For this stretch factor

$$y = f(3x)$$

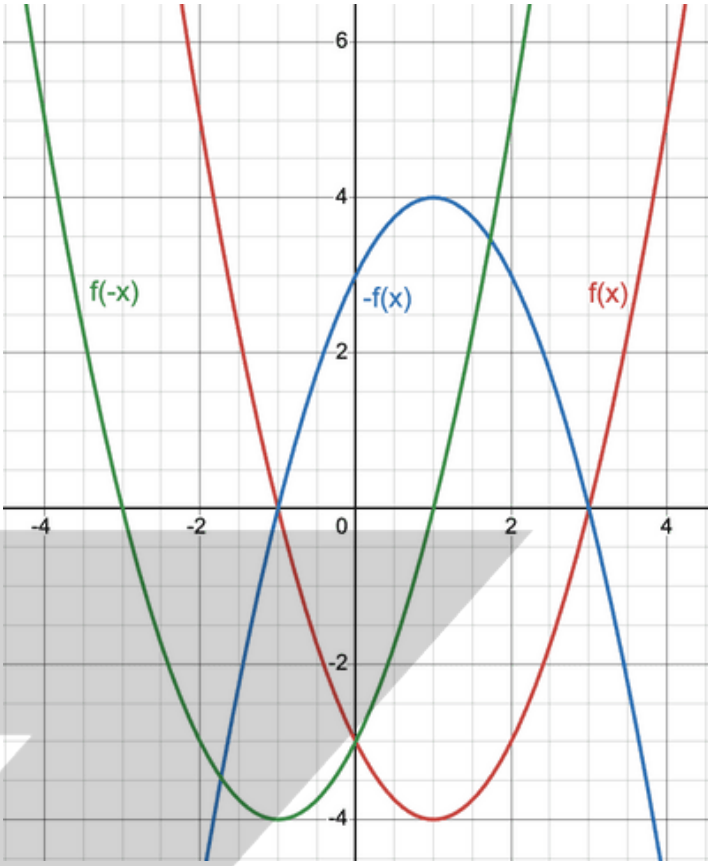
$$y = (3x)^2 + 3(3x) + 2$$

$$y = 9x^2 + 9x + 2$$

1.9. Reflections

- A reflection transformation reflects the graph to the x or y axis.

Function	Reflection	Result	Geometric interpretation	New points
$f(x)$	Reflection in the x-axis	$-f(x)$	Reflection in the x-axis. Graph is mirrored in the x-axis, which also acts as a line of symmetry to $f(x)$ and $-f(x)$.	$(x, -y)$
$f(x)$	Reflection in the y-axis	$f(-x)$	Reflection in the y-axis. Graph is mirrored in the y-axis, which also acts as a line of symmetry to $f(x)$ and $f(-x)$	$(-x, y)$



Example

Given the equation $f(x) = 2x^2 + 3x + 2$, find the resulting equations after a x and y axis reflection.

Answer

For a reflection in the y-axis

$$y = f(-x)$$

$$y = 2(-x)^2 + 3(-x) + 2$$

$$y = 2x^2 - 3x + 2$$

For a reflection in the x-axis

$$y = -f(x)$$

$$y = -(2x^2 + 3x + 2)$$

$$y = -2x^2 - 3x - 2$$

1.10. Order of Transformations

Order
Translation parallel to x-axis
Stretch parallel to x-axis
Reflection in the y-axis
Stretch parallel in y-axis
Reflection in the x-axis
Translation parallel to y-axis

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- To find out the transformations applied to a quadratic
  - Complete the square and then compare it with $f(x)$

Example

Functions f and g are both defined for $x \in \mathbb{R}$ and are given by

$$f(x) = (x-2)^2 + 5$$

$$g(x) = 2[(x+1)^2 + 5]$$

Describe fully the two transformations that have been combined to transform $f(x)$ to $g(x)$ and in the order they occur in.

Answer

The **first** transformation is the **translation** in the x-axis by 3 in the rightward direction. This can also be written as the vector $\binom{-3}{0}$.

$$f(x+3) = ((x+3)-2)^2 + 5 \implies f(x+3) = (x+1)^2 + 5$$

The **second** transformation is a **stretch** parallel to the y-axis by a stretch factor of 2.

$$2f(x+3) = 2[(x-2)^2 + 5]$$

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